Histology of the Testis and Sperm Release Pathway



Alexander S. R. Macnow, M.D., M.H.P.E.

Associate Professor, Division of Anatomy Education
Thread Director, Pathology
Department of Medical Education
Sidney Kimmel Medical College
Thomas Jefferson University

alexander.macnow@jefferson.edu

Questions? Concerns? Please don't hesitate to contact me by email at alexander.macnow@jefferson.edu.

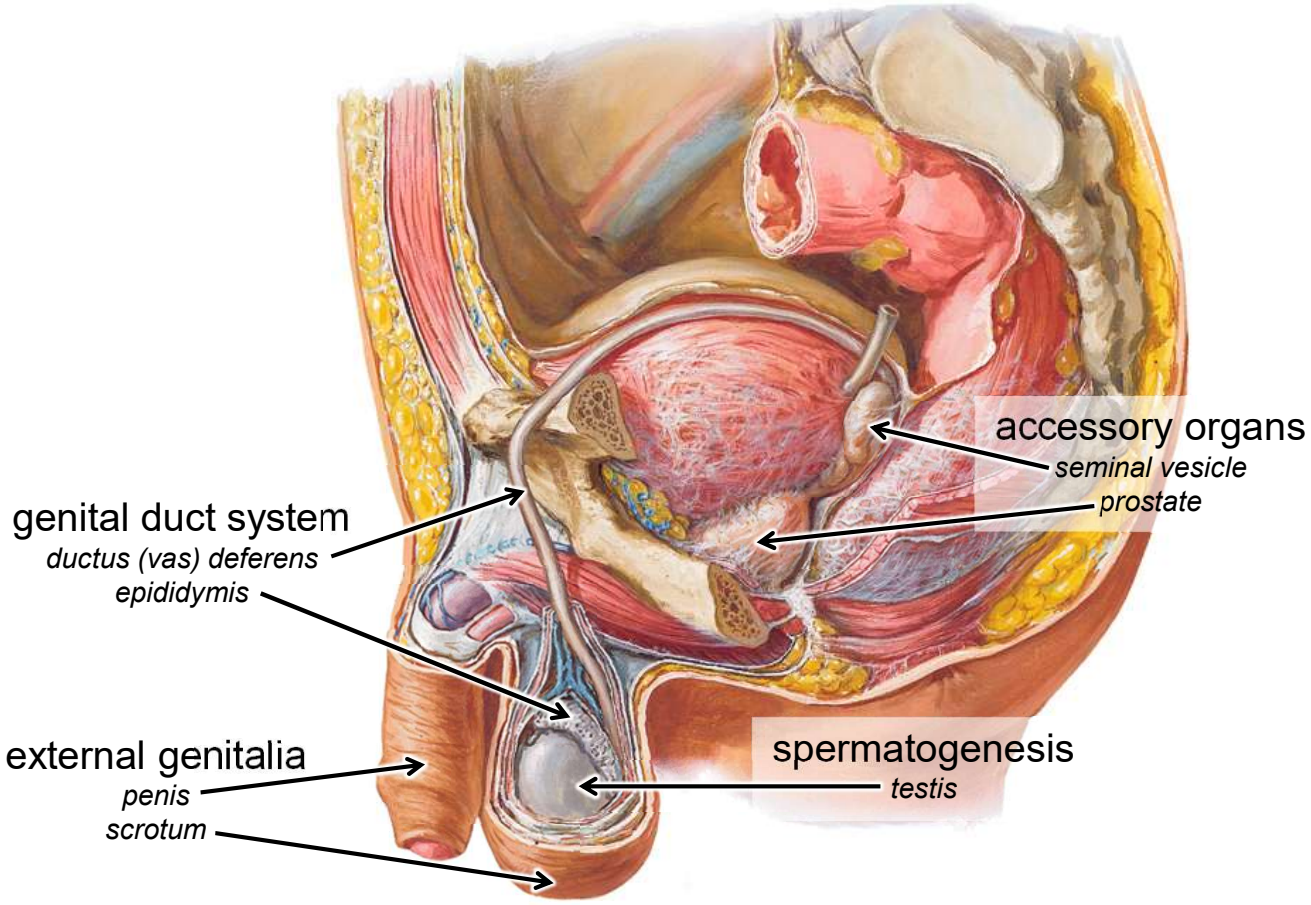
Image: An H&E-stained section of a lovely little heart-shaped prostatic gland containing some corpora amylacea.

LEARNING OBJECTIVES

By exam time, you should be able to do the following:

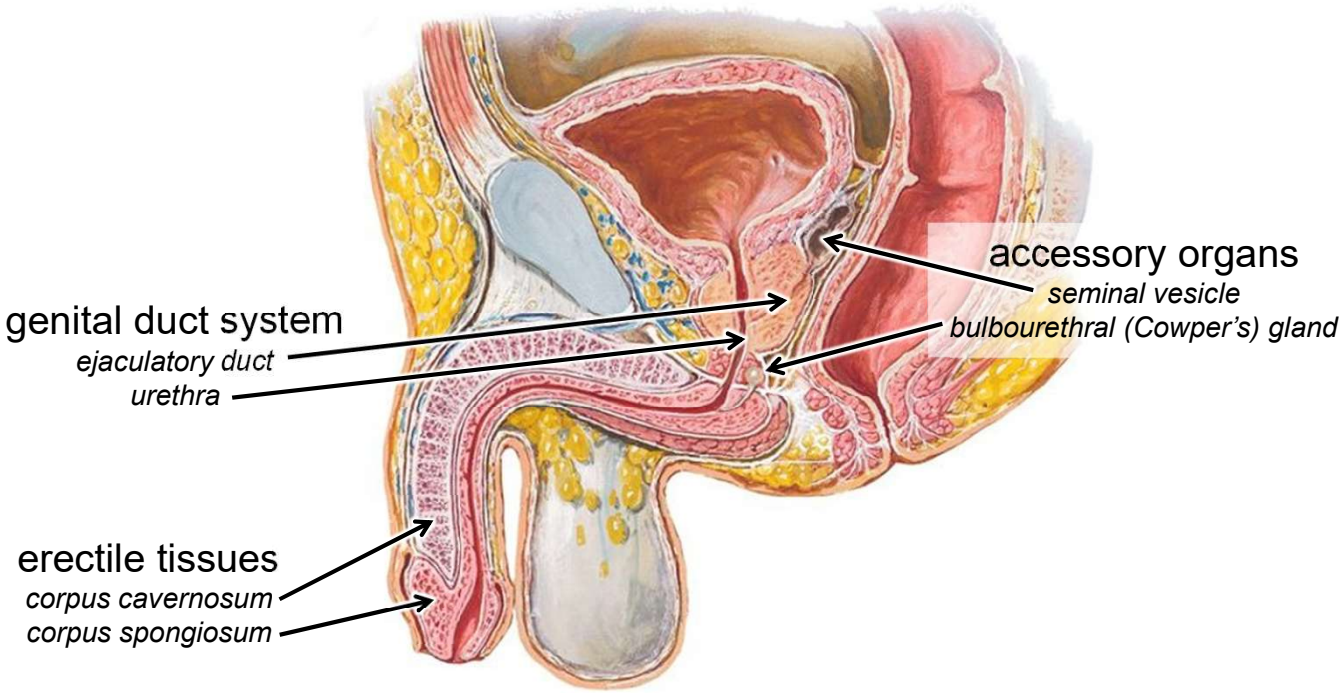
- Discuss the structural and functional anatomy of the testis and the organs involved in sperm maturation and release, including how they change over different life stages and the hormonal regulation of these changes.
- Describe and identify the cells and tissues of the testis and organs involved with sperm maturation and release by histology, including how those structures change over different life stages and via hormonal changes.
- Discuss the process of spermatogenesis from germ cell through mature sperm, including the different sites in the reproductive tract where the steps of maturation occur.
- Predict how different pathologies of the testis and organs involved in sperm maturation and release might alter the histological or physiologic features of the component cells/tissues.

GROSS ANATOMY: PARASAGITTAL VIEW



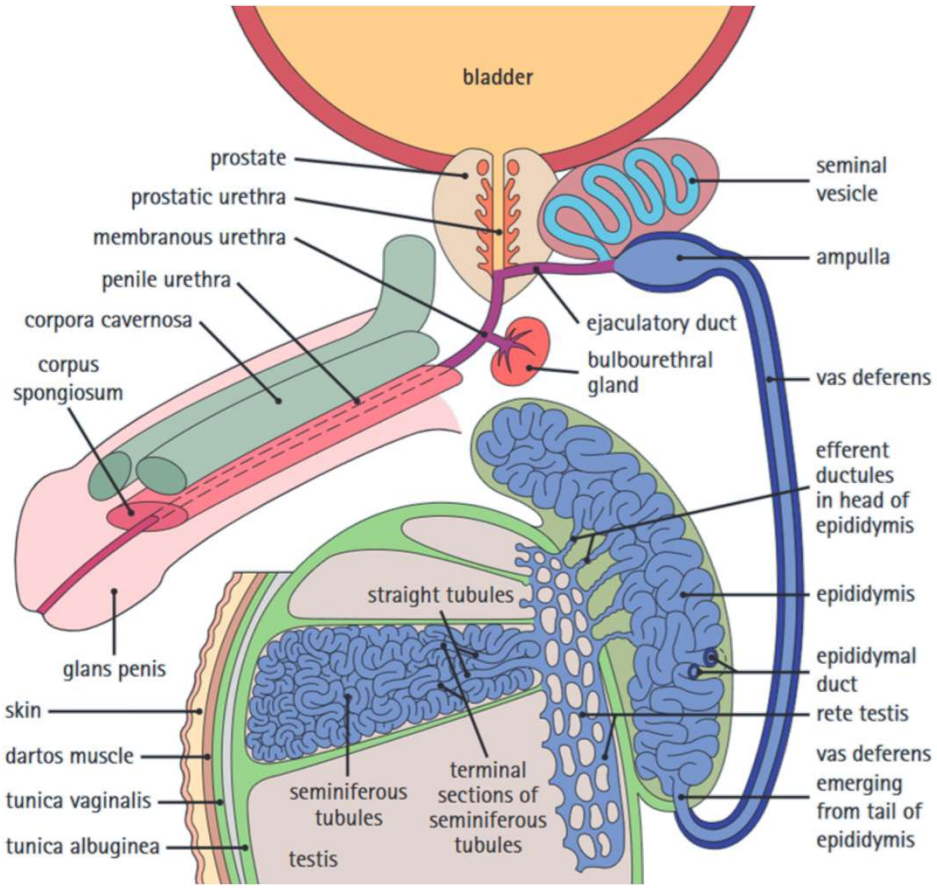
The testosterone-driven reproductive system contains several organs that can be classified based on function. **Spermatogenesis**, the production of haploid spermatozoa from diploid spermatogonia, occurs within the seminiferous tubules of the **testis** (**testicle**). These spermatozoa enter the genital duct system, grossly consisting of the **epididymis**, where they become motile and are stored until emission and ejaculation occur. During **emission**, spermatozoa continue from the tail of the epididymis into the **ductus** (**vas**) **deferens**, which carries them from the **scrotum** into the abdominopelvic cavity through the inguinal canal. The ductus continues internally to reach the posterior surface of the bladder, where it joins with the **seminal vesicle** to form the **ejaculatory duct**. The seminal vesicle represents one of three accessory organs; the **prostate**, which surrounds the urethra, constitutes another. After leaving the prostate, spermatozoa are transmitted through the urethra of the **penis**.

GROSS ANATOMY: SAGITTAL (MIDLINE) VIEW



In this sagittal view, we can see a few more structures that weren't well appreciated on the previous slide. We now can see the internal surface of the seminal vesicle and the **bulbourethral** (**Cowper's**) **gland** embedded in the deep perineal pouch, alongside the external urethral sphincter. Within the prostate, the **ejaculatory duct** (formed when the secretions of the seminal vesicles join with the ductus deferens) can be seen cutting through the substance of the gland to bring seminal fluid to the **urethra**. Here, the urethra is a common pathway for urine and seminal fluid; when vulva and a vagina are present, the reproductive tract and urethra have separate openings into the vestibule. Within the penis, the two types of erectile tissue are demonstrated: there are two **corpora cavernosa** in the dorsal aspect of the penis, and a single **corpus spongiosum** in the ventral aspect. The penile (spongy) urethra passes through the corpus spongiosum to reach the glans penis and empty into the outside world.

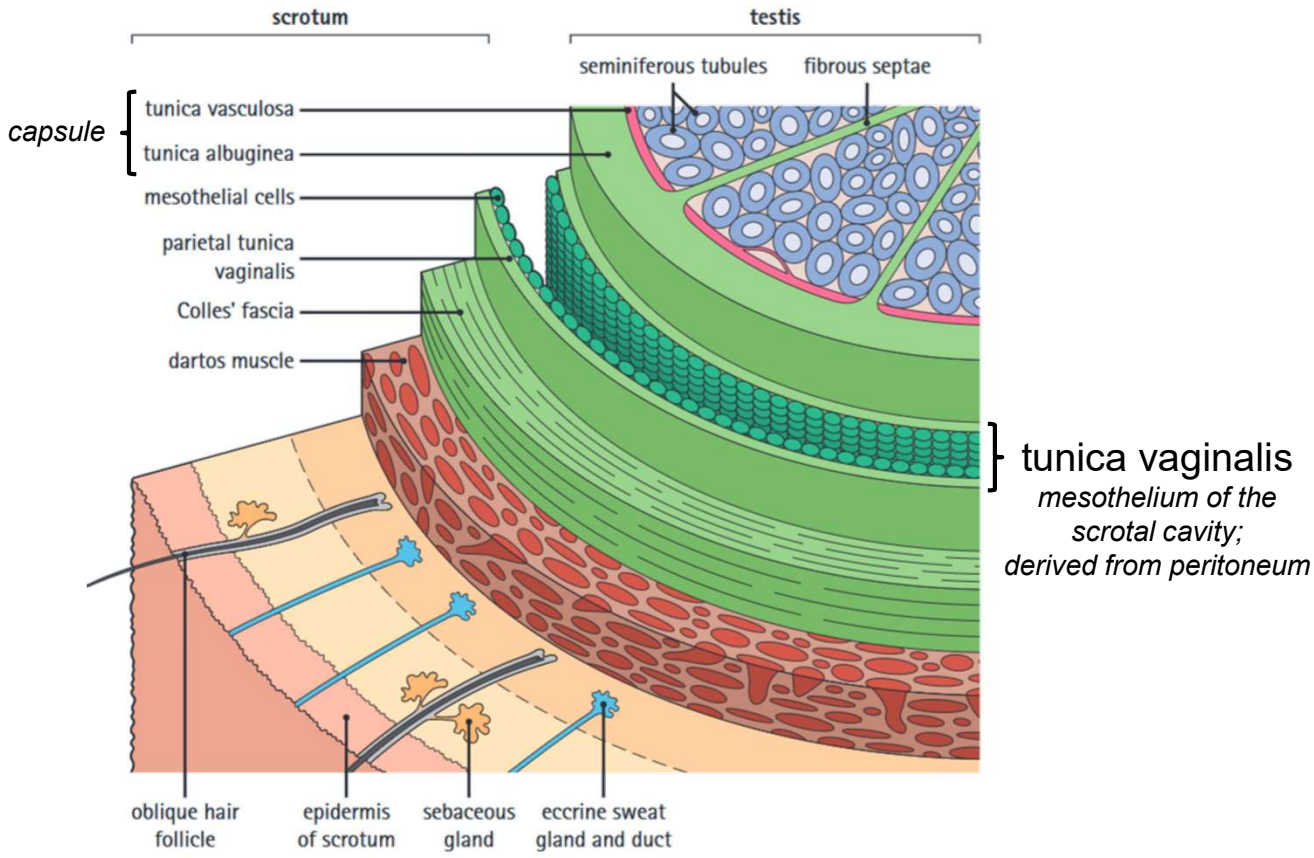
SCHEMATIC ORGANIZATION OF SPERM PATHWAY



This image schematizes the pathway we'll be looking at during the remainder of this lab. While the structures are not shown to scale, it shows the interrelationships between their lumina and is a useful organizational tool.

Testis

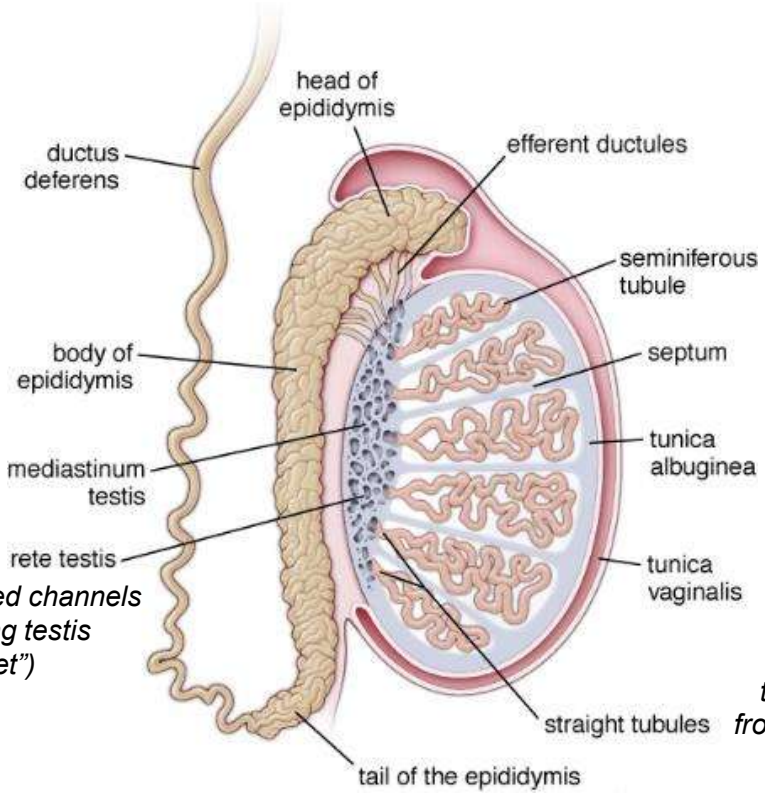
STRUCTURE OF SCROTAL CONTENTS



The **testes** are bilateral structures located in the **scrotum**, outside the coelom (body cavity). They migrated prenatally from the retroperitoneum into this location by traversing the inguinal canal. Their position outside the body cavity is preferable for spermatogenesis, which occurs best approximately 2–3 °C below body temperature. They are both exocrine glands, as the seminiferous tubules produce spermatozoa; and endocrine glands, as the interstitial **Leydig cells** and the intratubular **Sertoli cells**, produce **testosterone** and **inhibin**, respectively. The testis is enclosed in a thick capsule, the **tunica albuginea**, and septations extend from the tunica into the testis, dividing it into approximately 250 to 300 lobules.

The scrotum consists of hair-bearing skin overlying a layer of smooth muscle, the **dartos muscle**. Deep to the dartos muscle is **Colles's fascia**, a fibrocollagenous layer continuous with the Colles's fascia of the perineum (and Scarpa's fascia of the abdomen); its deepest portion is a condensation that represents the **parietal layer of the tunica vaginalis**, the mesothelium of the scrotal cavity. Like the other three mesothelia we've learned about—the serous pericardium, pleura, and peritoneum—it's a simple squamous layer of epithelium producing a trace amount of fluid to allow for reduced friction with movement. In this case, that's movement of the testis up and down within the scrotum in response to temperature and other stimuli. The tunica vaginalis "doubles back" on itself at the superior and inferior aspect of the scrotal cavity to cover the testis and head of the epididymis as the **visceral layer of the tunica vaginalis**. The exterior capsule of the testis itself is composed of a fibrocollagenous **tunica albuginea** (*alb–* = "white") overlying a thinner, vasculature-containing **tunica vasculosa**.

ORGANIZATION OF THE INTRATESTICULAR TUBULES



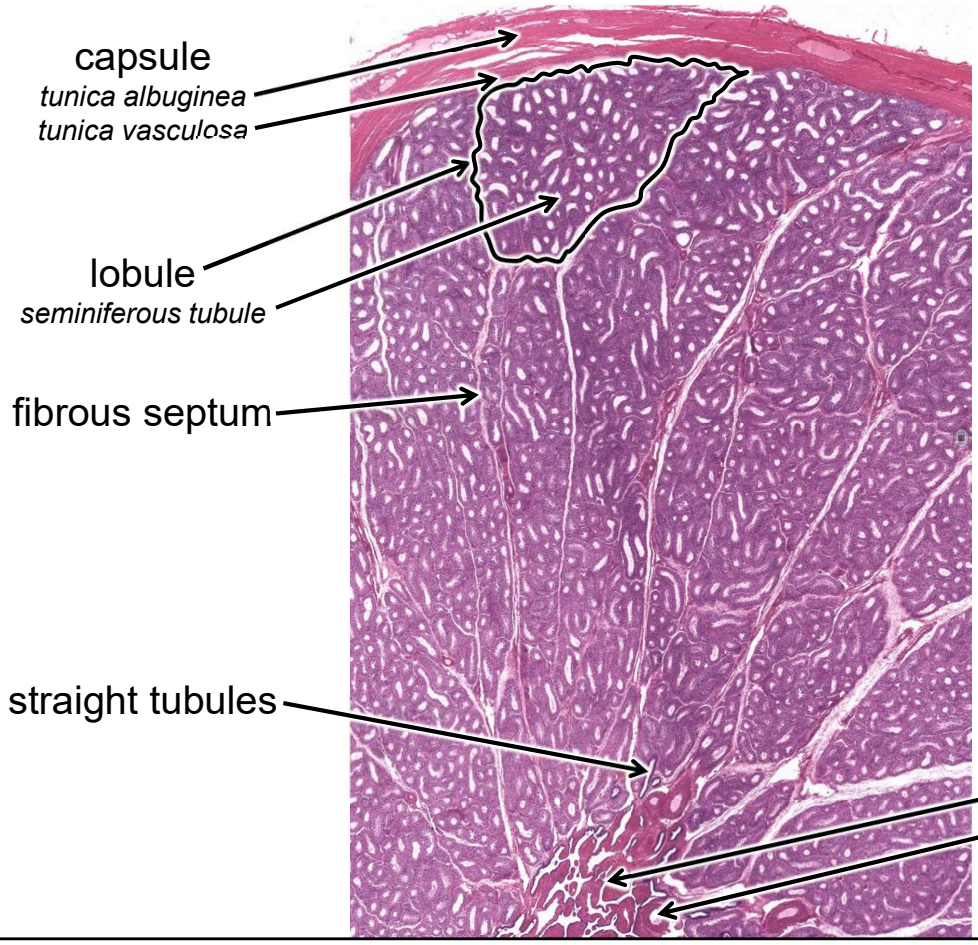
organized into ~250 pyramid-shaped lobules; 1–4 "U"-shaped tubules per lobule

large, interconnected channels for sperm exiting testis (–ret– = "net")

transmit spermatozoa from seminiferous tubules to rete testis

This parasagittal cross-section shows the relationships of the testis, tunica vaginalis (with visceral and parietal layers), epididymis, and ductus (vas) deferens. Our focus, however, will be on the organization of the testis. The testis is divided into approximately 250–300 pyramid-shaped **lobules**, each of which contains one to four **seminiferous tubules**. The seminiferous tubules are U-shaped, with their ends oriented toward the **mediastinum testis** (essentially the hilum of the organ). These seminiferous tubules drain into **straight tubules**, which transmit developed but immotile spermatozoa to the **rete testis** within the mediastinum. The **rete testis** (*ret–* = "net"), a system of interconnected, anastomosing channels that drain the spermatozoa toward the epididymis.

HISTOLOGY OF THE TESTIS: LOW POWER

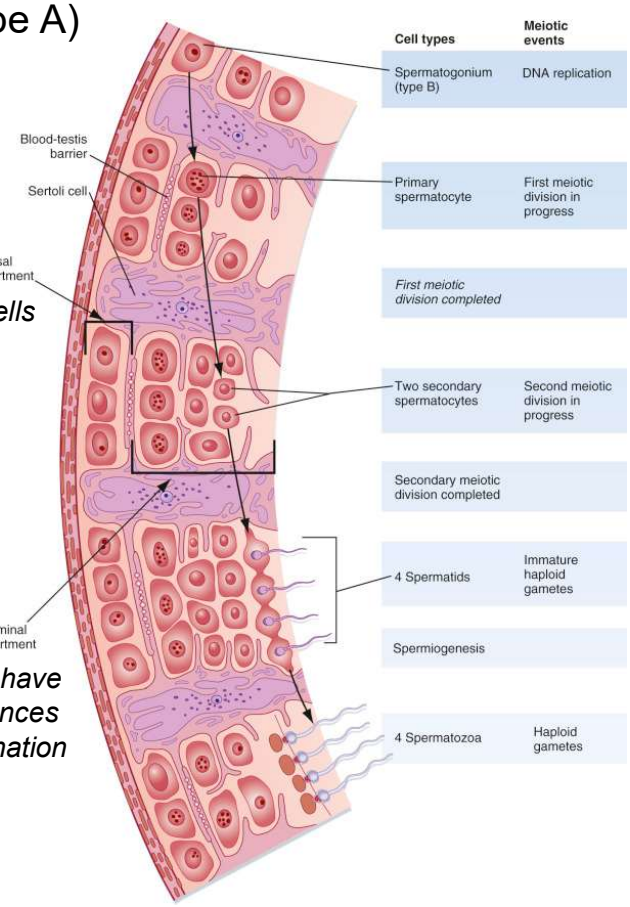


At low magnification, we can get a sense of the organization of the testis on this H&E-stained slide. At the very top, a small portion of the **tunica albuginea** is present. The **tunica vasculosa** is less apparent but may be represented by a small, longitudinally-cut vessel in this specimen. Note that the majority of the image is comprised of tightly packed profiles of tubules with thick walls and variable shapes: these are the **seminiferous tubules**. The variability of the profiles and their proximity to each other indicate that the seminiferous tubules are closely packed and highly coiled. Most nuclei present in the tubules belong to the **germinal epithelium**, where they represent several stages of development of **sperm** (**spermatozoa**): the small, tail-bearing gametes. The seminiferous tubules are organized into lobules, divided by fibrous septa. Toward the bottom of the image, we reach the **mediastinum testis**, containing the **rete testis** and vasculature supplying the organ. The mediastinum represents the hilum of the organ (like that of the lung, kidney, spleen, liver [as porta hepatis], etc.): the region where vasculature, innervation, lymphatics, and other relevant tubular structures enter and exit the organ.

SPERMATOGENESIS

not pictured: spermatogonium (type A)

undergoes mitosis to maintain pool
(type B is committed to meiosis)



contains diploid germ cells

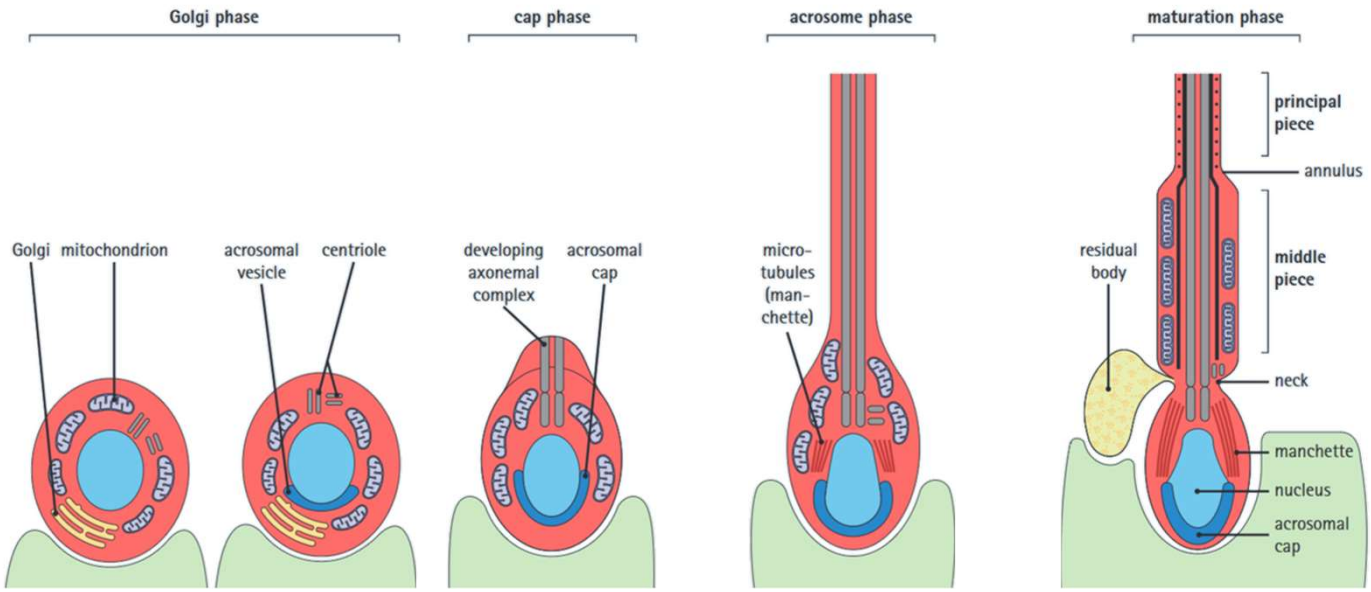
contains meiotic cells which have novel ("foreign") DNA sequences and proteins due to recombination

Spermatogenesis is the process by which stem cells, the **spermatogonia**, are transformed into mature **spermatozoa** (**sperm**). This process begins shortly before puberty. Spermatogonia may be **type A**, which undergo mitosis to replenish the stem cell pool (auto-renewal), or **type B**, which are committed to undergoing meiosis and gametogenesis. Type A dark spermatogonia have dark nuclei and divide frequently to maintain the stem cell population. Type A pale spermatogonia have paler nuclei and divide by mitosis several time but then mature into type B. Type B spermatogonia have clumped chromatin.

The type B spermatogonia enter **meiosis I** through the usual features (chromosomal condensation, dissolution of membrane-bound organelles, etc.). Once in **prophase I**, these are diploid **primary spermatocytes**. Starting at this stage (primary spermatocyte), all meiotic development must occur beyond the **blood–testis barrier**, formed by tight junctions between Sertoli cells. This barrier is necessary because the meiotic cells will have undergone recombination (crossing-over) and will express novel proteins that may be interpreted as "foreign" by the immune system. After undergoing meiosis I, the resulting haploid **secondary spermatocytes** quickly undergo **meiosis II** to become **early spermatids** (also haploid). These early spermatids elongate and condense their nuclei to become **late spermatids**, which are released from the Sertoli cells. Finally, these late spermatids undergo **spermiogenesis** to form **spermatozoa**, which flow toward the straight tubules and rete testis.

SPERMIOGENESIS

formation of mature (though immotile) spermatozoa from spermatids



polarization: Golgi toward basal end, centriole toward apical end; early acrosome formation

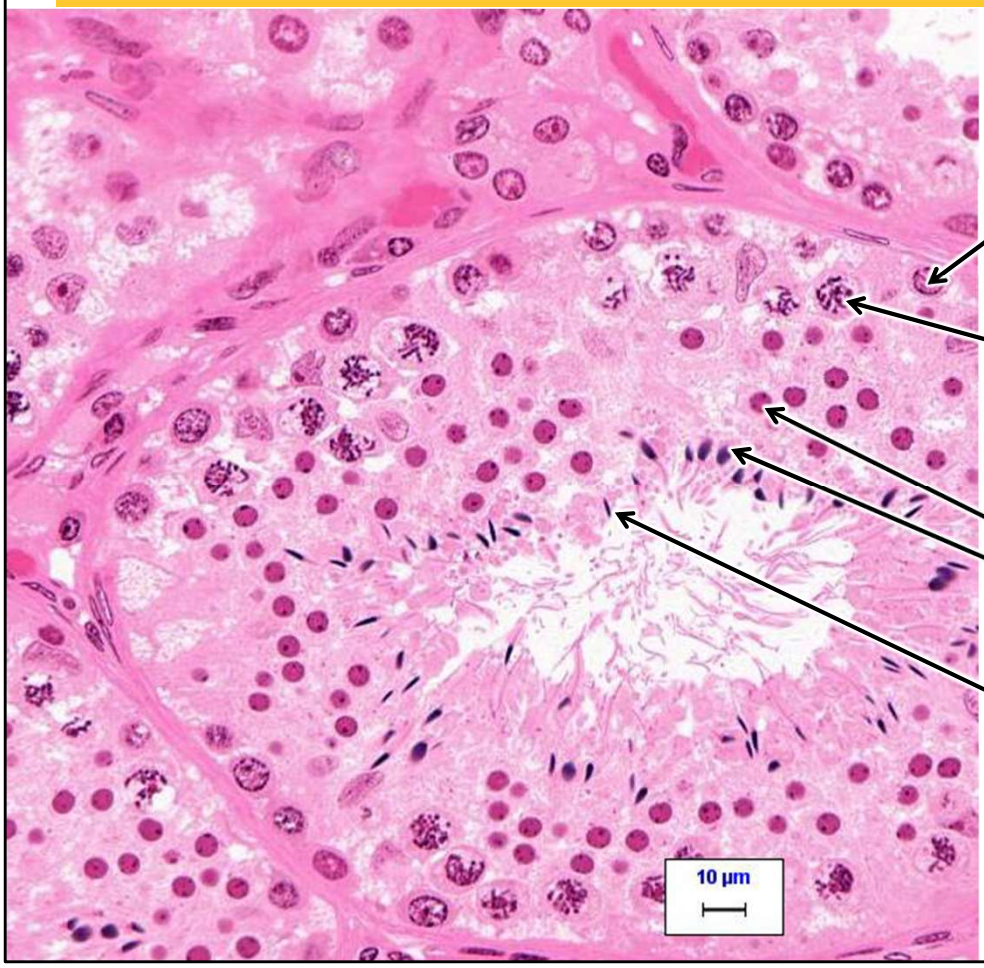
acrosome covers anterior half of nucleus; DNA condensation

nucleus flattens and elongates; mitochondria migrate toward developing middle piece; tail developed from centriole

excess cytoplasm and organelles pinched off as residual body (phagocytosed by Sertoli cells); releases from Sertoli cell surface

The transformation of the spermatids into spermatozoa, a phased process known as **spermiogenesis**, involves elongation of the nucleus, formation of an **acrosome** by the Golgi complex, migration of centrioles and formation of a flagellum from the distal centriole, regrouping of the mitochondria around the base of the flagellum and decrease in cytoplasm through the extrusion of a **residual body** (which is phagocytosed by the Sertoli cells).

HISTOLOGY OF SPERMATOGENSIS: HIGH POWER

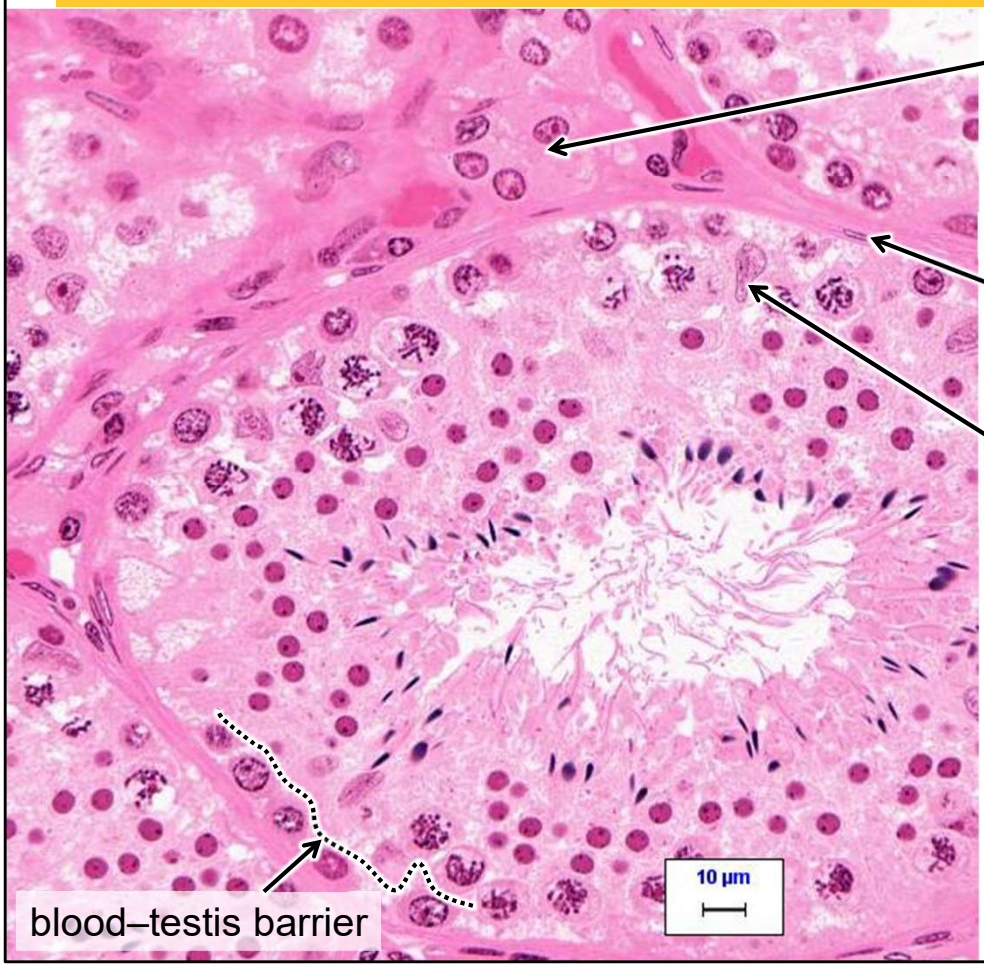


While evaluating the histology of the **germinal epithelium**, three criteria are useful in finding the stages of sperm cell differentiation:

1. Cells in the same stage are grouped together;
2. The longer the time span of a developmental stage, the more frequently are cells in that stage seen; and
3. Earlier stages are located basally and later stages apically in the tubule wall.

Spermatogonia are found adjacent to the basement membrane in the basal compartment of the seminiferous tubule. **Primary spermatocytes** have large nuclei with condensed chromosomes and are located immediately on the luminal side of the blood–testis barrier formed by the Sertoli cells. Primary spermatocytes spend a long time in the various phases of the first meiotic division, and thus they are frequently found. The **secondary spermatocytes** are smaller, have less prominent chromatin and go through the second meiotic division very quickly, so they are rarely found. The **early spermatids**, the final result of the two meiotic divisions, are found in large clusters near the lumen of the tubules. They have a small round nucleus with finely granular chromatin and occasionally some dense chromatin mass. Then, the **late spermatids**, with small dark, exclamation point-shaped nuclei, are found at the luminal surface of the tubules, with their tails in the lumen. Note that, despite having acquired their final morphology, the spermatozoa are still immotile at this point in development.

OTHER CELLS OF THE TESTICULAR LOBULE

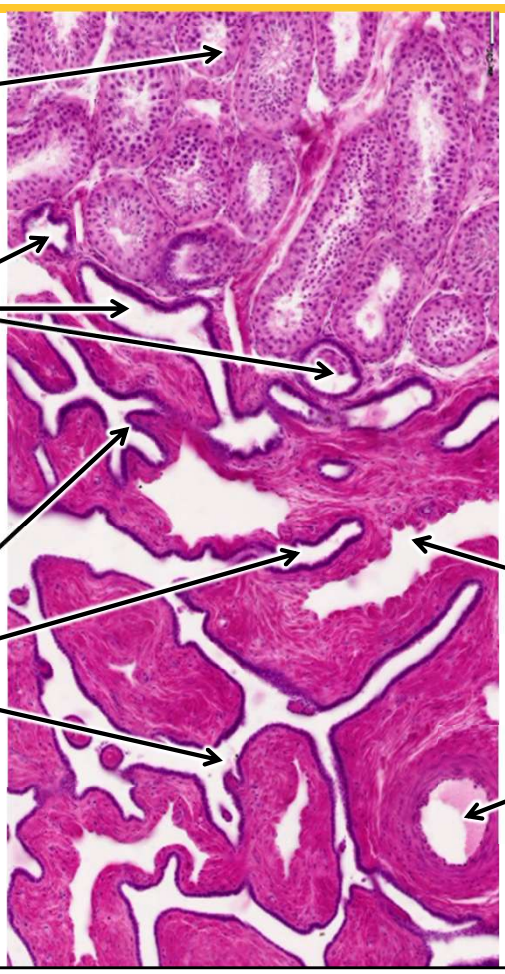


At high magnification, note the **myoid cells** surrounding each seminiferous tubule. These cells are contractile in nature, propelling the spermatozoa toward the ends of the seminiferous tubules; at this point, the spermatozoa are still immotile. There is a thin basement membrane dividing each seminiferous tubule from its related myoid cells, though this is difficult to see directly in these images.

Sertoli cells comprise only about 10% of the cells in the epithelium of the adult (post-pubertal) seminiferous tubule. Look for a fairly large, pale nucleus with a nucleolus located in the basal third of the tubule wall. The most distinctive characteristic is that the nucleus is elongated in the radial direction, while other nuclei are round. Some of the cytoplasm of the Sertoli cell is visible, but most is difficult to see because the cytoplasmic processes of these cells are interposed between the developing sperm cells. Sertoli cells respond the **follicle-stimulating hormone** (**FSH**), and are responsible for supporting and regulating sperm development, as described earlier. They are also part of the **hypothalamic–pituitary–testis** (**HPT**) **axis**, secreting **inhibin** and **activin** to feed back to the central nervous system. They secrete **androgen binding proteins** (**ABP**) to concentrate androgens in the luminal region of the tubule. During development, these cells produce **anti-Mullerian hormone** (**AMH**), which causes regression of the Mullerian duct system (which otherwise forms the oviduct, uterus, cervix, and upper one-third of the vagina). The Sertoli cells are also phagocytic, resorbing **residual bodies** from the meiotic process as well as faulty spermatozoa. They are also responsible for forming and maintaining the **blood–testis barrier**, also as described earlier.

Finally, **interstitial** (**Leydig**) **cells** are large, polygonal cells that produce androgens, including **testosterone**, **androstenedione**, and **dehydroepiandrosterone** (**DHEA**). Their cytoplasm may appear somewhat foamy due to the cholesterol precursors needed for steroidogenesis. In prenatal life, these hormones drive development of the testis, penis, and duct system. At puberty, they drive sexual maturation and the development of secondary sex characteristics (facial and chest hair, laryngeal enlargement, increased muscle mass, increased hair coarseness). In adulthood, they maintain sperm quality and duct patency. These cells are close to capillaries, attesting to their endocrine function. Leydig cells also produce **oxytocin**, which drives peristalsis in the seminiferous tubules.

HISTOLOGY OF MEDIASTINUM TESTIS



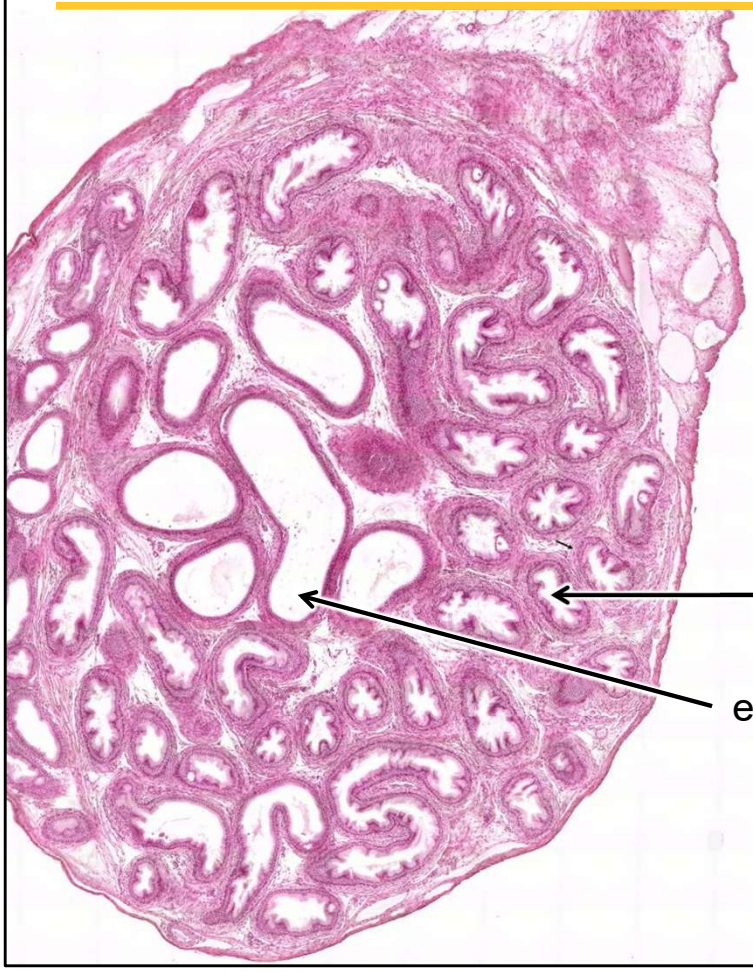
As spermatozoa leave the **seminiferous tubules**, they enter the **straight tubules**: short segments with only Sertoli cells and no germ cells, which connect the terminal ends of the U-shaped seminiferous tubules to the rete testis. The **rete testis** represents the channels that conduct maturing spermatozoa from the seminiferous tubules to the epididymis. The channels of the rete testis are lined by simple cuboidal epithelium and are derived from seminiferous tubules but do not have either Sertoli cells or spermatogonia. They lead to **efferent ductules** and the **epididymis**, both of **mesonephric** (**Wolffian**) **duct** origin.

Also present in the mediastinum testis, alongside the rete testis, are arteries and veins supplying the organ. Arteries are generally easy to identify from their well-organized smooth muscle walls. Veins may be easier to confuse with the rete testis; however, the rete testis is lined with simple cuboidal epithelium, whereas veins have a simple squamous epithelium.

Genital Duct System: Efferent Ductules, Epididymis and Ductus (Vas) Deferens

The genital duct system is responsible for carrying spermatozoa from their production in the testis until they reach the outside world. This system consists of the efferent ductules of the testis, epididymis, ductus (vas) deferens, ejaculatory duct, and urethra. In this section, we'll consider the first three of those; we'll look at the urethra at the end of this pre-lab.

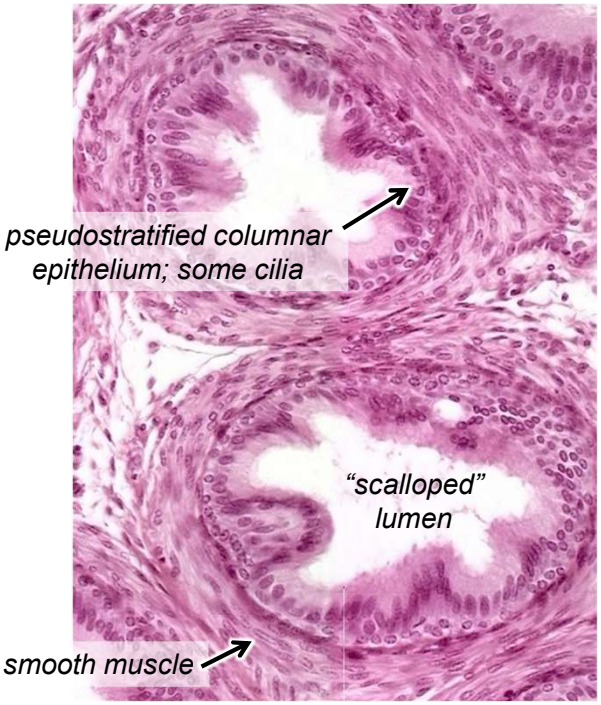
HISTOLOGY OF THE EPIDIDYMIS: LOW POWER



The organ grossly identified as the **epididymis** contains both the **epididymis proper** (**ductus epididymis**) as well as the **efferent ductules** that connect from the rete testis to the epididymis proper.

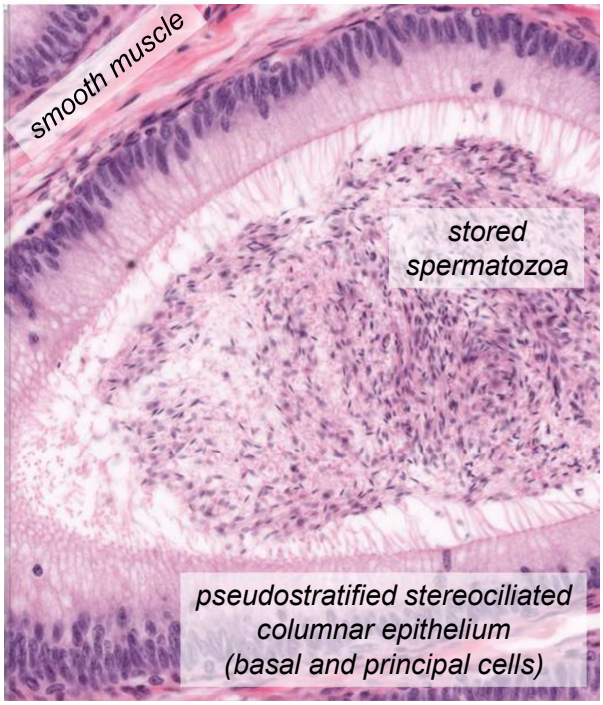
As an organ, the epididymis is responsible for accumulating, maturing, and storing spermatozoa. Part of the maturation process in this location is the development of flagellar motility; however, note that most of the movement of spermatozoa through the genital duct system (including during ejaculation) is due to muscular contraction, not flagellar activity. Further maturation includes the stabilization of condensed chromatin, changes in the plasma membrane surface charge, and acquisition of new surface proteins (particularly those acting as binding proteins for the zona pellucida (the "shell" of the oocyte). The cells lining the epididymis have **stereocilia** (branched microvilli), allowing them to resorb significant amounts of testicular fluid used to propel the spermatozoa out of the testis. These cells can also phagocytose abnormal spermatozoa, serving as a quality control. Finally, the epididymal cells secrete several factors that keep the spermatozoa "in check" until they need to be active, including **immobilin**, a massive glycoprotein with viscoelastic properties that significantly reduce flagellar movement (and reduce energy waste), and other **sperm decapacitating factors** that prevent premature activation of the acrosome reaction. Spermatozoa are stored in the tail of the epididymis, near where it meets the ductus (vas) deferens.

HISTOLOGY OF THE EPIDIDYMIS: HIGH POWER



efferent ductules

drain from rete testis to head (superior pole) of epididymis



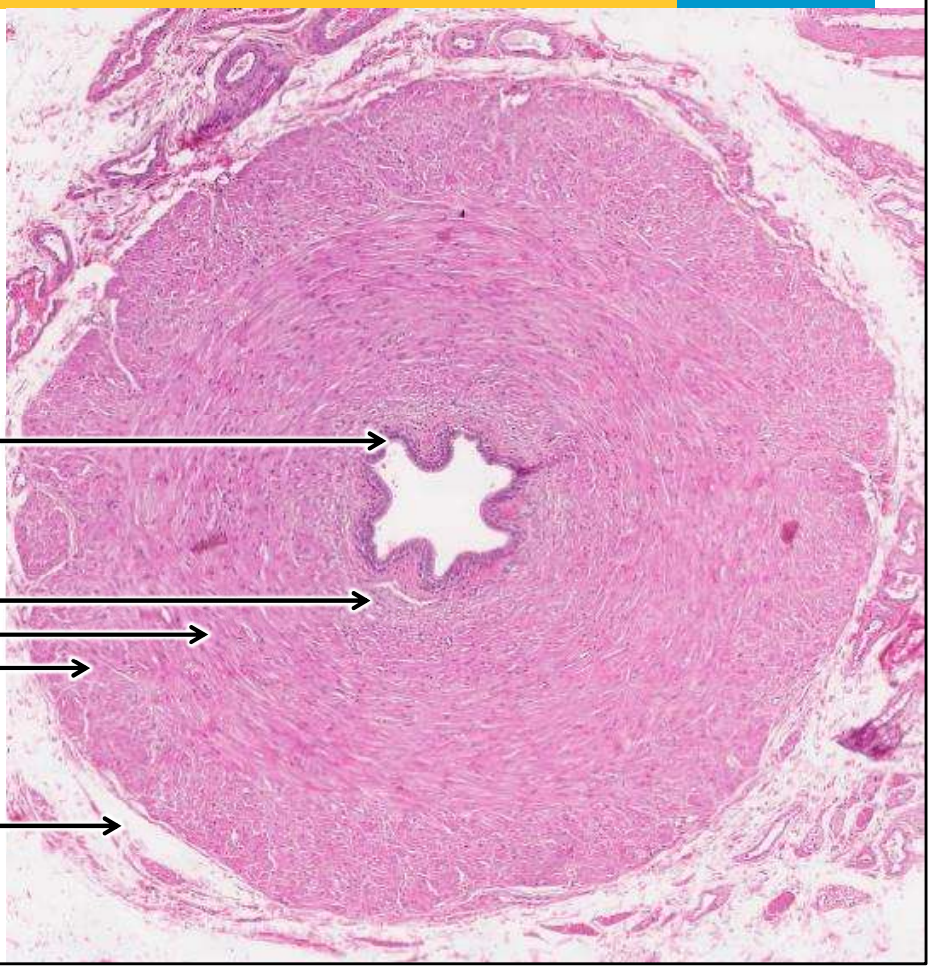
epididymis proper

single large tube draining to ductus deferens; gross: head (superior), body, and tail (inferior); tail stores spermatozoa until ejaculation

The **efferent ductules** that carry spermatozoa and testicular fluid from the rete testis to the **head** of the **epididymis** have a characteristic "scalloped" morphology, with a crowded, pseudostratified columnar epithelium without some cilia but no stereocilia (see below). The ductules are surrounded by a coat of smooth muscle.

The **epididymis** proper is lined with pseudostratified columnar epithelium, in which the small basal cells are thought to be regenerative and precursors of the tall **principal cells**. The apical surfaces of these tall cells contain extremely long branching microvilli, (inappropriately) known as **stereocilia**, as they are non-motile, contain actin filaments (not microtubules), and have no basal bodies. The epithelial cells lie on a basement membrane surrounded by a circular layer of smooth muscle. The sperm are stored in great abundance in the distal portion of the epididymis (the **tail**), where the ionic, viscoelastic medium keeps them immotile in order to conserve their energy stores.

HISTOLOGY OF THE DUCTUS (VAS) DEFERENS



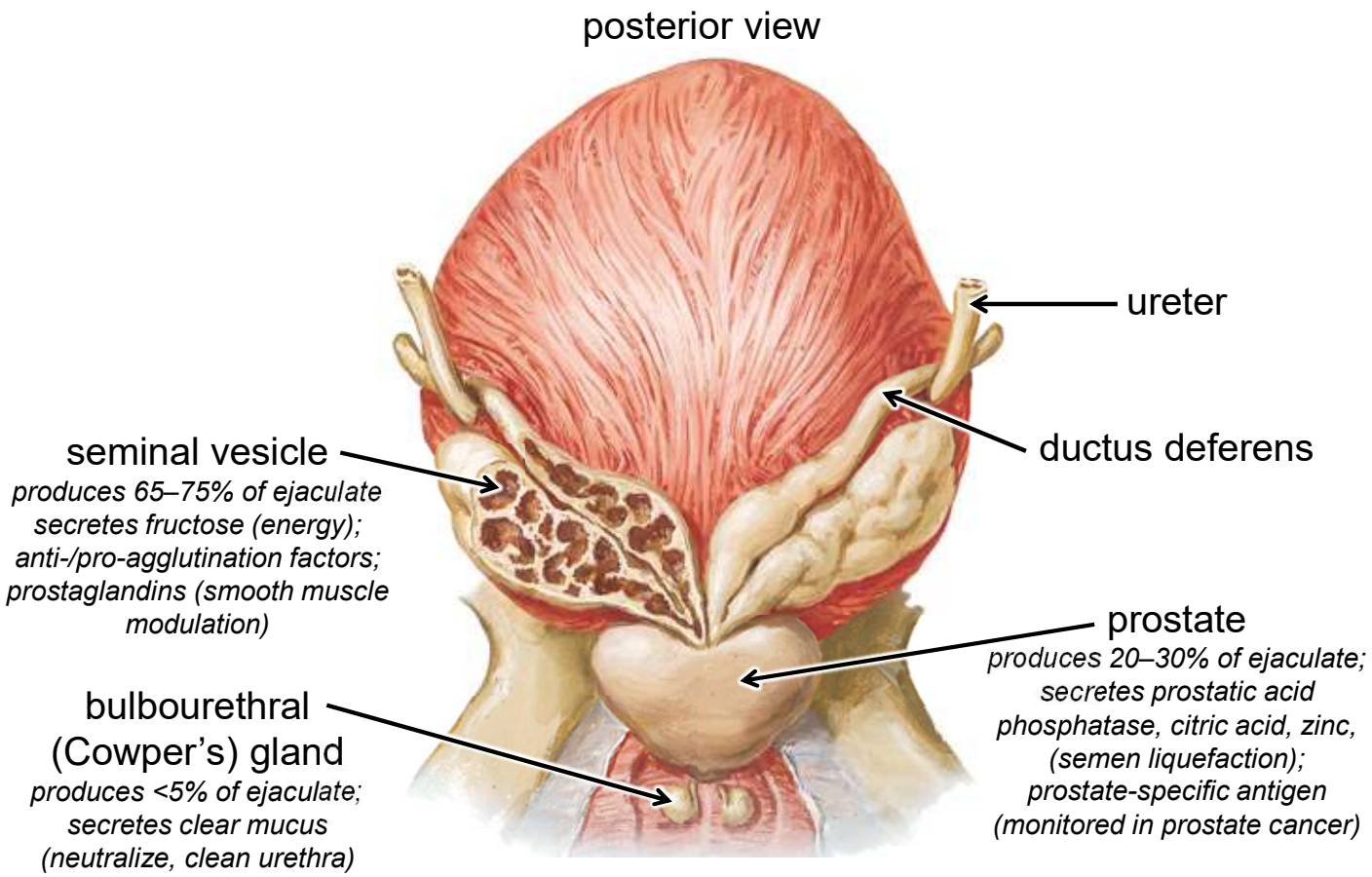
The **ductus** (or **vas**) **deferens** is a continuation of the epididymis. It leaves the scrotum and passes through the inguinal canal as a component of the spermatic cord, which includes the vas deferens, the testicular artery, and the pampiniform plexus of testicular veins. Behind the urinary bladder, it joins the **seminal vesicle** to form the **ejaculatory duct**, which pierces the prostate and opens into the prostatic urethra. Just before reaching the prostate gland the ductus deferens is dilated to form the **ampulla**. The vas is the structure that is ligated and partially resected during vasectomy.

The layers consist of a pseudostratified, partially stereociliated columnar epithelium overlying three layers of smooth muscle: an inner longitudinal (unusual!) layer, an intermediate circular layer, and an outer longitudinal layer. The layers become thinner at the ampulla, which may serve as a reservoir for sperm between emission (movement of the spermatozoa from the epididymis) and ejaculation (rhythmic propulsion of seminal fluid).

Finally, the ductus deferens is surrounded by an **adventitia**, which contains collagenous connective tissue to anchor the organ in place as well as blood vessels, nerves, and lymphatics that supply the organ.

Accessory Organs

ACCESSORY GLANDS: OVERVIEW



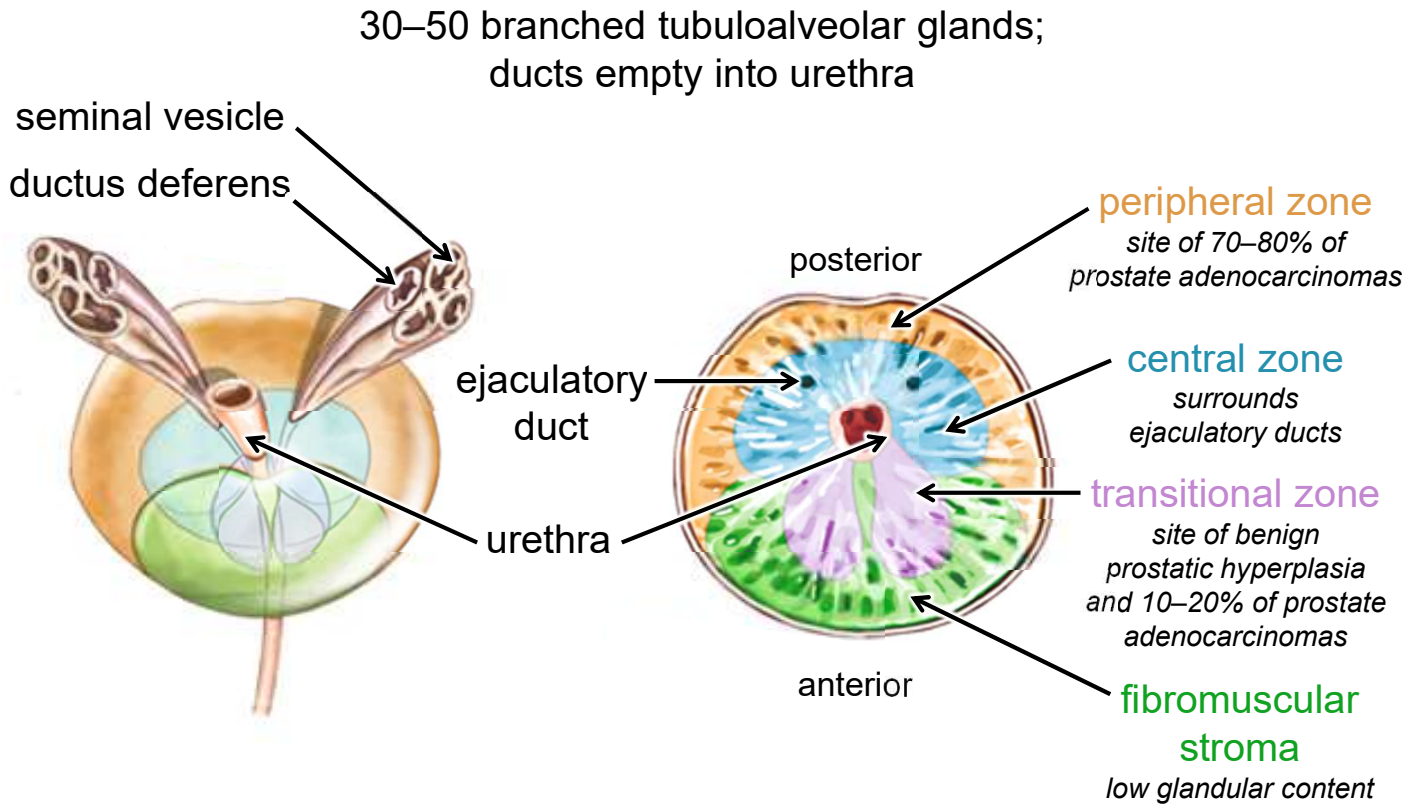
The testosterone-driven reproductive tract has three major accessory glands: the seminal vesicles, the prostate gland, and the bulbourethral glands. In this lab, we'll look at the histology of the first two of those: the seminal vesicles and the prostate gland.

The **seminal vesicles** are elongated, coiled saccular structures lying posterior to the prostate gland, which join the vasa deferentia to form the ejaculatory ducts before the latter pass through the prostate to the urethra. The epithelial cells secrete fructose-rich fluid, providing a source of energy to the spermatozoa. They also produce anti-agglutination factors to prevent sperm sticking together within the genital duct system, but also pro-coagulative factors (like seminogelins) that cause seminal fluid to thicken in viscosity upon release. They also modulate smooth muscle activity through prostaglandins.

The **prostate gland** produces most of the remainder of the seminal fluid, about 20–30%. (Note that spermatozoa account for only ~1% of seminal volume.) It secretes prostatic acid phosphatase, which reverses the activity of seminogelins to cause thinning of seminal viscosity. This process is citric acid and zinc-dependent. In the context of reproduction, these viscosity changes initially allow retention of semen in the vaginal canal and its subsequent thinning and mobilization to migrate through the uterine cervix. The prostate also produces **prostate-specific antigen**, which is used (with some controversy) in the workup of prostatic disease and monitoring of prostate cancer.

The **bulbourethral** (**Cowper's**) **gland** serves most of its function *before* ejaculation, producing mid-viscosity, translucent mucus (**pre-ejaculate**). These secretions are released during erection and emission, prior to ejaculation, and serve to neutralize the urethral pH and protect the seminal fluid (given that the urethra is a common pathway for urine, too).

REGIONS OF THE PROSTATE GLAND

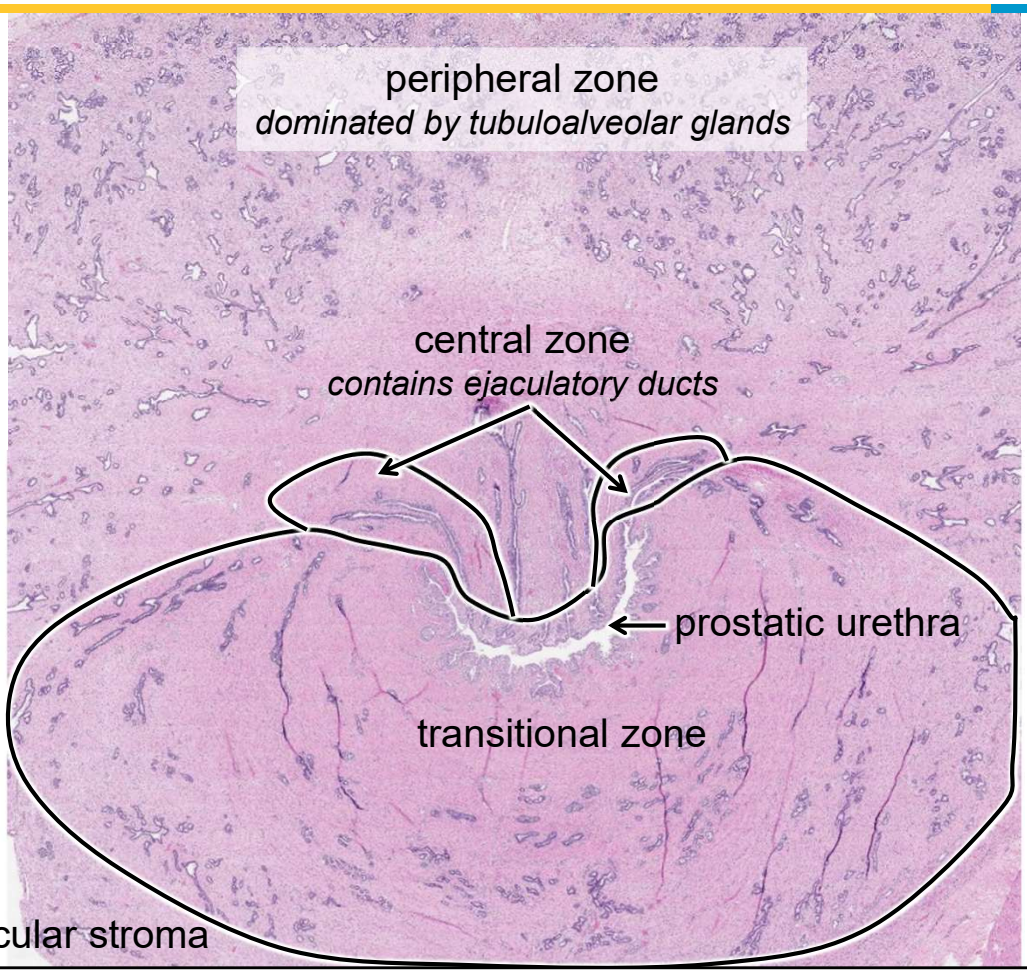


The **prostate gland** is a branched, compound gland that surrounds the proximal part of the urethra. Entering into its posterior aspect are the shared lumina of the seminal vesicles and ductus deferentes, which merge on each side to form **ejaculatory ducts** on each side. The ejaculatory ducts drain into the **prostatic urethra**.

The prostate is divided into four regions:

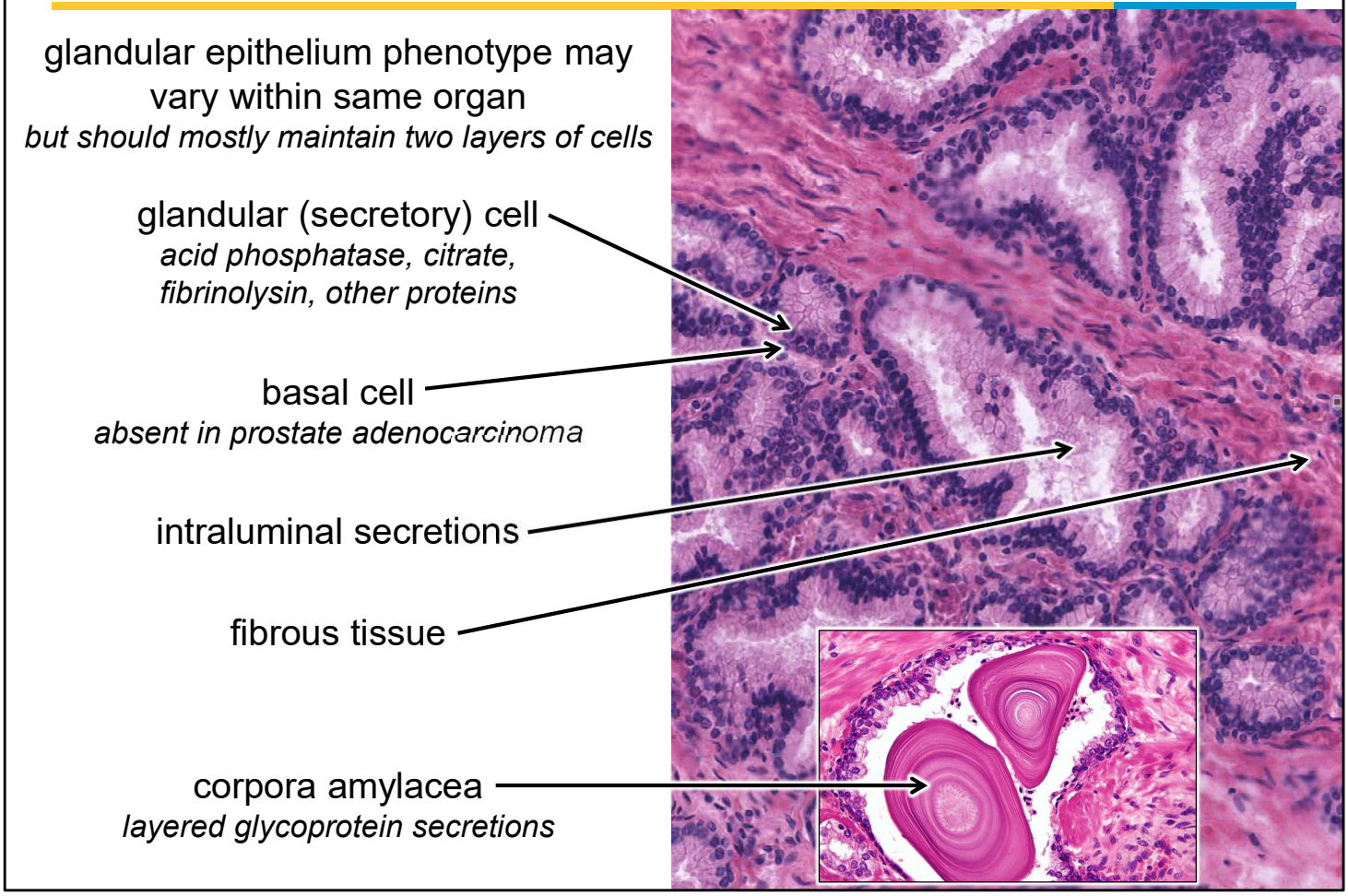
- The **peripheral zone** occupies most of the posterolateral portions of the gland and is rich in glandular tissue. It is the predominant site of prostate adenocarcinomas, accounting for approximately 70–80% of cases. Fortunately, this portion of the prostate is easily palpated on a digital rectal examination.
- The **central zone** surrounds the ejaculatory ducts and cuts obliquely through the gland.
- The **transitional zone** is in proximity with, and also surrounds, the prostatic urethra. It is the region affected by benign prostatic hyperplasia (BPH) and accounts for most remaining cases of prostate adenocarcinoma (approximately 10–20% of cases). Its proximity to the urethra explains the urinary symptoms experienced by those with BPH.
- The (**anterior**) **fibromuscular stroma** is a gland-poor region toward the anterior aspect of the gland.

HISTOLOGY OF THE PROSTATE GLAND: LOW POWER



The prostate gland is composed of tubuloalveolar glands that empty into the lumen of the urethra by way of about 20 excretory ducts. The prostatic urethra is seen in the center, with a U-shaped configuration (it may also appear Y-shaped on these histological sections, owing to its collapsed appearance when urine is not flowing). It is surrounded by the **transitional zone**. The terminal ends of the ejaculatory ducts are also seen emptying into the urethra, allowing us to identify the **central zone**. Toward the top of the slide is the gland-rich **peripheral zone**, while the gland-poor **fibromuscular stroma** is seen toward the bottom.

HISTOLOGY OF THE PROSTATE GLAND: HIGH POWER



The prostate is comprised of alkaline mucus-secreting glands and ducts that ultimately empty into the urethra. Each gland is surrounded by a small rim of connective tissue, and larger bands of fibrous connective tissue divide the gland overall into lobules. The epithelium contains two layers: a luminal layer of secretory cells that take on variable appearance. They may be columnar (as seen here) to cuboidal to even squamous, depending on how active they are. Regardless, there is an underlying layer of small, cuboidal basal cells. The presence of this second cell population is critical: when absent, it indicates that the glands are malignant. The lumina of the acini may contain abundant secretions. In some cases, these secretions may form layered structures called **corpora amylacea** ("starchy bodies") that may calcify over time.

One additional thought from a pathologist: one of the biggest challenges with prostate pathology is that the histological differences between normal prostate and prostate cancer actually are the "reverse" from most other organs: normal prostate tissue has very complex, very blue glands. Prostate cancer, in contrast, tends to have simple, small, often pale-staining neoplastic glands (at least for Gleason grade 3). This can make it rather challenging, especially for less experienced trainees, to pick up on cancers percolating in the background.

HISTOLOGY OF THE SEMINAL VESICLES: LOW POWER

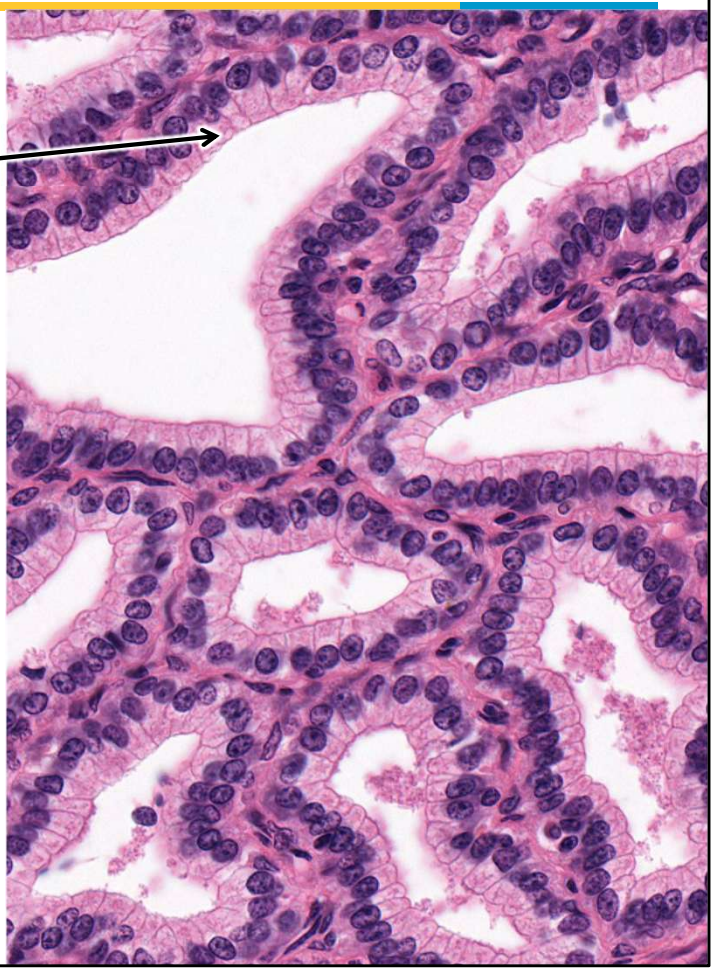


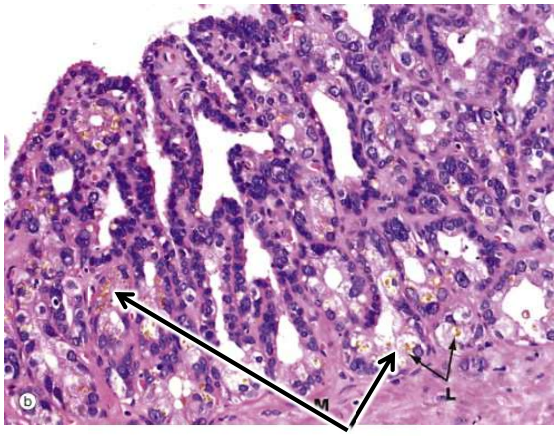
The most striking histologic characteristic of the **seminal vesicle** is the folding of its mucosa. This forms a complex appearance with honeycomb-like architecture. In glands with a more apparent lumen, the mucosa may take on a more villous appearance toward the center. The mucosa consists of pseudostratified columnar epithelium resting on cellular connective tissue that is surrounded by two layers of smooth muscle, an inner circular layer and an outer longitudinal layer. Note that the lumen of the seminal vesicle does not ordinarily contain spermatozoa.

HISTOLOGY OF THE SEMINAL VESICLES: HIGH POWER

secretory epithelium

pseudostratified to simple columnar;
secretes fructose (energy);
anti-/pro-agglutination factors; prostaglandins (smooth muscle modulation)





may see lipofuscin granules

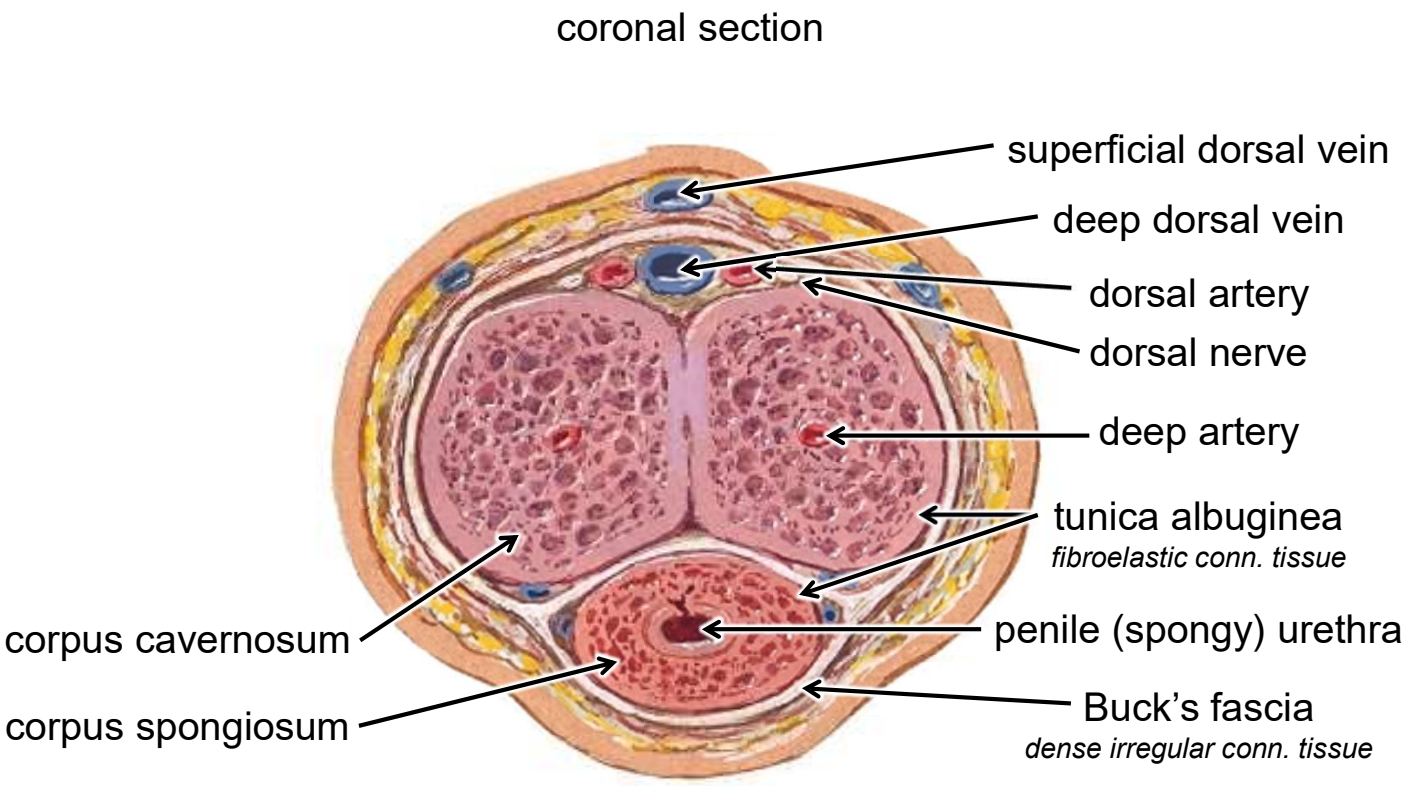
granular, yellow-brown "wear and tear" byproduct

At higher power, the pseudostratified epithelium of the seminal vesicle is more easily seen. In less crowded specimens, the epithelium may be more simple columnar in appearance. The epithelial cells secrete fructose-rich fluid, providing a source of energy to the spermatozoa. They also produce anti-agglutination factors to prevent sperm sticking together within the genital duct system, but also pro-coagulative factors (like seminogelins) that cause seminal fluid to thicken in viscosity upon release. They also modulate smooth muscle activity through prostaglandins.

In some specimens, especially those of older individuals, it is common to see granules of **lipofuscin**, a yellow-brown granular intracellular accumulation that represents complexes of lipids and proteins due to cell membrane damage, usually due to free radicals (highly reactive molecules with unpaired electrons). Accordingly, lipofuscin is often thought of as a representation of "wear-and-tear."

External Genitalia

STRUCTURE OF THE PENIS

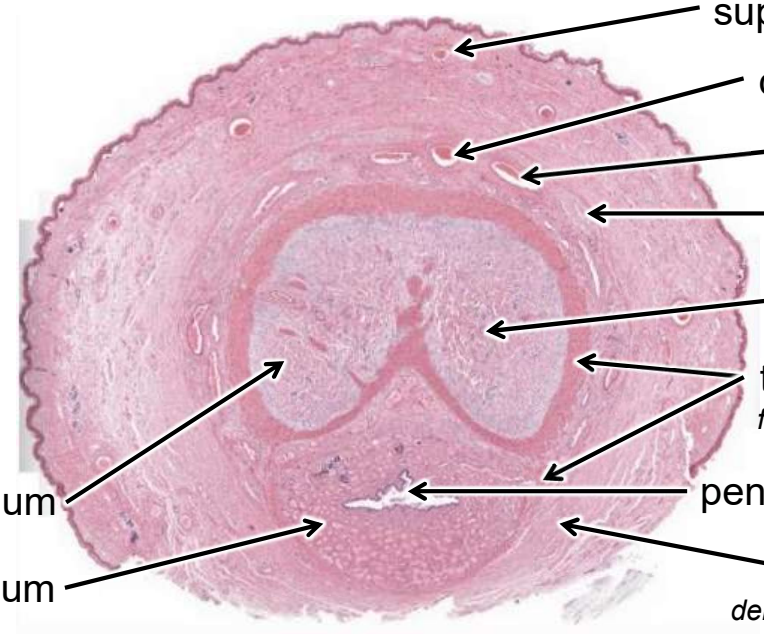


The **penis** contains chambers of erectile tissues, which contain interconnected vascular sinuses that are empty when the penis is flaccid and filled when the penis is erect. At the dorsal aspect of the penis are two **corpora cavernosa**, which contain these vascular sinuses. The main blood supply into these corpora cavernosa are the **deep** (**cavernosal**, **central**) **arteries**. At the ventral aspect is the **corpus spongiosum**, with less rigid vascular sinuses and containing the **penile** (**spongy**) **urethra**. The penile urethra is a common pathway for spermatozoa as well as urine. At the periphery of each chamber of erectile tissue is a thickened "rind" of fibroelastic connective tissue, the **tunica albuginea** (same name as the capsule of the testis). The chambers overall, as well as some of their associated circulatory and nervous structures, are contained within the **Buck's fascia** (corresponding to this region's deep fascia). It is composed of dense irregular connective tissue.

Within the Buck's fascia are the midline **deep dorsal vein**, flanked on each side by the **dorsal arteries** and the **dorsal nerves**. Superficial to Buck's fascia are cutaneous veins, including the midline **superficial dorsal vein**.

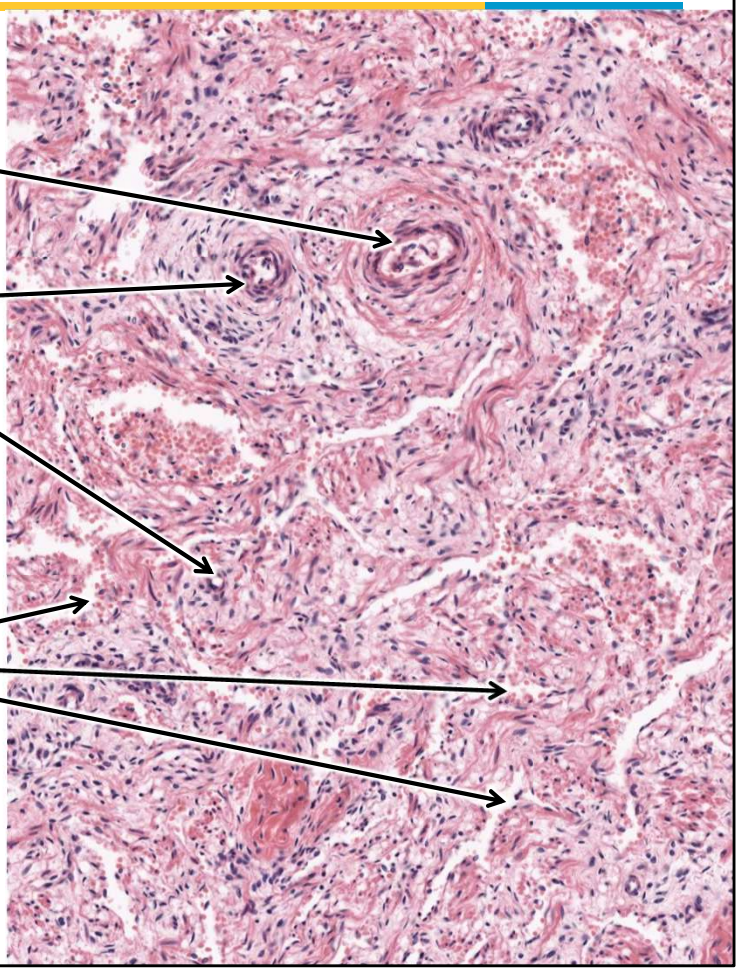
HISTOLOGY OF THE PENIS: LOW POWER

coronal section



In this whole-mount coronal section of the penis, we simply correlate the structures on the previous slide with their histological counterparts.

HISTOLOGY OF ERECTILE TISSUES: CORPUS CAVERNOSUM

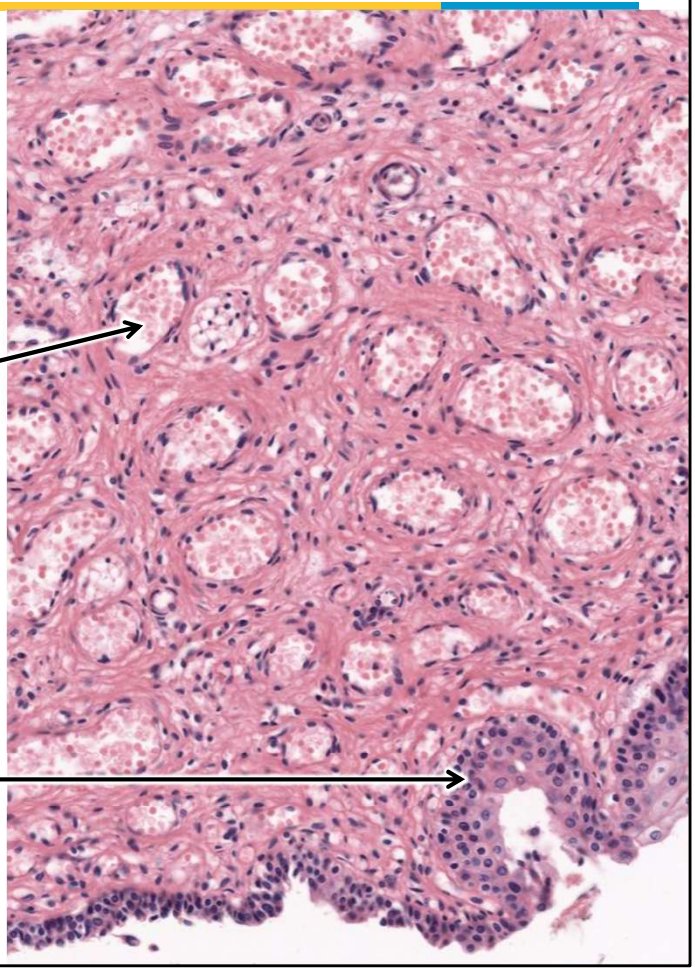


Let's catalogue microcirculation through the erectile tissues. In the corpora cavernosa, arterial blood flows in through **deep** (also called **cavernosal** or **central**) arteries: one in each chamber. Branching from the deep arteries are labyrinthine **helicine arteries**, which empty into endothelium-lined **vascular sinusoids**. The walls of these sinusoids contain smooth muscle cells and fibrous connective tissue. When erection is initiated, the parasympathetic outflow from the pelvic splanchnic nerves (as **cavernosal nerves**, within the chamber), triggers the release of nitric oxide, which activates guanylyl cyclase and produces cyclic GMP (cGMP), relaxing smooth muscle in the arterial wall and causing vasodilation. This increased flow to the erectile tissues causes the sinusoids to fill and pressure increases within the corpus cavernosum. The tunica albuginea serves as a relatively rigid wall, leading to turgor of the external genitalia. This also compresses venous outflow from the corpus cavernosum until a steady state is reached at the increased intracorporal pressure. This pressure is further maintained by the "clamping" action of the ischiocavernosus muscles, which cover the proximal portions of the corpora cavernosa as they arise from the pelvic ischia. Note that the above description applies to both the corpora cavernosa of the penis as well as those of the clitoris.

Tying over to Pharmacology, phosphodiesterase type 5 (PDE5) inhibitors block the breakdown of cyclic GMP (cGMP), maintaining vasodilation and increased flow into the corpus cavernosum. These drugs are used for erectile dysfunction, and end with the suffix "*–afil*" (e.g., sildenafil, tadalafil, vardenafil).

Another clinical correlate: blockage of the outflow of the vascular sinusoids may lead to **priapism**: a prolonged, persistent erection. This may be seen in sickle cell disease and other structural hemoglobinopathies, as a side effect of certain medications (classically, trazodone, a treatment for depression and insomnia), or with the use of erectile dysfunction drugs (often off-label or not prescribed, including the PDE5 inhibitors above or injectable preparations). Priapism is a urologic emergency; sustained blockage of venous drainage can lead to infarction of the tissue.

HISTOLOGY OF ERECTILE TISSUES: CORPUS SPONGIOSUM



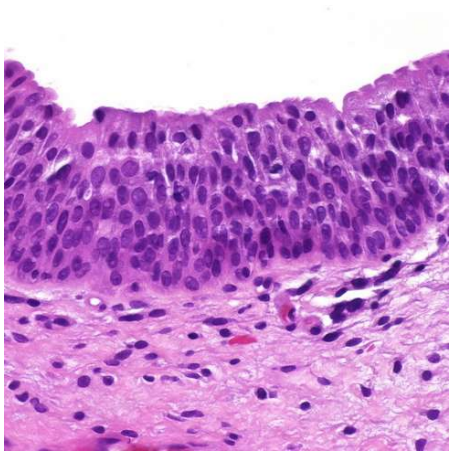
vascular stroma
thinner fibromuscular walls

urothelium
transitional epithelium
only in penis
(clitoral corpora spongiosa lay deep to labia minora and surround vestibule, not urethra)

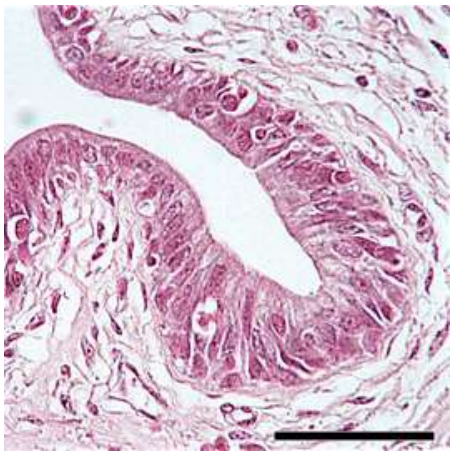
The erectile tissue of the **corpus spongiosum** is similar, with vascular sinusoids surrounded by fibromuscular walls. However, the walls surrounding each sinusoid are thinner than in the corpus cavernosum. This is true for both the single corpus spongiosum of the penis, as well as the bilateral corpora spongiosa of the clitoris.

In those with penises, the corpus spongiosum will also surround the **penile** (**spongy**) **urethra**, which is lined by **urothelium** (transitional epithelium).

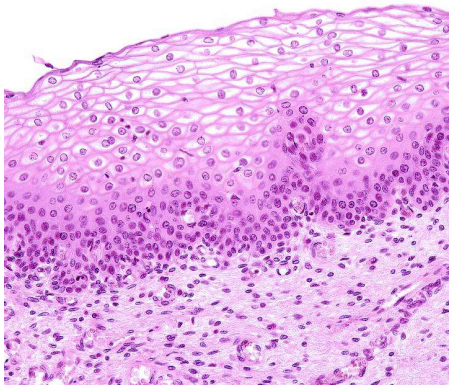
EPITHELIA OF THE URETHRA



transitional (urothelium)



pseudostratified columnar



stratified squamous

when prostate and penis are present:

prostatic urethra

membranous urethra;
penile (spongy) urethra

terminal end of penile (spongy) urethra

when vagina is present, and urethra opens to vestibule:

proximal two-thirds

distal one-third

The **urethra** has variable epithelia, depending on where one is along its course. We introduced these epithelia back in in the Renal block but are revisiting them here.

When the prostate and penis are present, the urethra takes on the following epithelia as listed below:

- At the beginning of the urethra, where it passes through the prostate (**prostatic urethra**), it is lined by transitional epithelium (urothelium).
- As it passes from the prostate toward and into the penis (**membranous urethra** and **spongy urethra**), it is lined by pseudostratified columnar epithelium.
- As it reaches the end of the spongy urethra, it becomes lined by stratified squamous epithelium.

When the urethra opens independently into the vestibule, above the vagina, its proximal two-thirds or so have transitional epithelium (urothelium), and its distal one-third has stratified squamous epithelium.

LEARNING OBJECTIVES

By exam time, you should be able to do the following:

- Discuss the structural and functional anatomy of the testis and the organs involved in sperm maturation and release, including how they change over different life stages and the hormonal regulation of these changes.
- Describe and identify the cells and tissues of the testis and organs involved with sperm maturation and release by histology, including how those structures change over different life stages and via hormonal changes.
- Discuss the process of spermatogenesis from germ cell through mature sperm, including the different sites in the reproductive tract where the steps of maturation occur.
- Predict how different pathologies of the testis and organs involved in sperm maturation and release might alter the histological or physiologic features of the component cells/tissues.

Histology of the Testis and Sperm Release Pathway



Alexander S. R. Macnow, M.D., M.H.P.E.

Associate Professor, Division of Anatomy Education
Thread Director, Pathology
Department of Medical Education
Sidney Kimmel Medical College
Thomas Jefferson University

alexander.macnow@jefferson.edu

Questions? Concerns? Please don't hesitate to contact me by email at alexander.macnow@jefferson.edu.

Image: An H&E-stained section of a lovely little heart-shaped prostatic gland containing some corpora amylacea.